Statement of Research Experience & Interests

1. Overview & Immunoengineering Vision

My research program centers on the rational design, chemical synthesis, and supramolecular assembly of **lipid-based and polymeric biomaterials** engineered to interface with biological membranes and direct cellular outcomes. Across **7+ years of research experience** (producing **9 published peer-reviewed papers** and **6 manuscripts submitted/in preparation**), I have built an integrated toolkit spanning controlled polymerization, bioconjugation, liposome/lipid-nanoparticle formulation, and multi-scale biophysical characterization. Joining the Deak Laboratory on your newly funded NIH R01 projects represents the ideal convergence of my synthetic and lipid-nanoparticle expertise with translational **immunoengineering**—specifically developing **liposomal nanotrap therapeutic HIV vaccines** and **tolerogenic dendritic cell biomaterials**.

2. Chemical Synthesis, Bioconjugation & Surface Functionalization

Controlling how biomaterials interact with immune cells requires precise chemical control over polymer architecture, surface charge, and ligand presentation:

- **Controlled RAFT Polymerization & Block Architectures:** I established kinetic and synthetic rules for alternating and block vinyl ether–maleic anhydride (VEMA) copolymers via RAFT polymerization ($M_n = 6\text{–}40 kDa$, $Đ \sim 1.3\text{–}1.5$; *Biomacromolecules* 2024; *Polymer* 2026) and chain-extended macro-CTAs with hydrophobic acrylate blocks (*Polymer Chemistry* 2026) to tune amphiphilic balance and self-assembly.
- **Modular Bioconjugation (PEG, Amino Acids & Charge Tuning):** Utilizing the reactive maleic anhydride backbone as a versatile chemical handle, I executed nucleophilic ring-opening bioconjugations with **polyethylene glycol (PEG)** chains, **amino acids**, and anionic, cationic, and zwitterionic modifiers to create **9 charge-tunable derivatives (VEMADs)**. These surface-functionalized systems maintain colloidal stability across **pH 3.0–9.0** and up to **30 mM** Mg^{2+}. In the Deak Lab, I will directly apply this bioconjugation chemistry to tether targeting peptides, antigens, and immunomodulatory ligands onto liposomal and polymeric scaffolds.

3. Liposomal, Lipid-Based & Polymeric Biomaterials for Viral & Biological Targets

My experimental core involves formulating and characterizing liposomes, proteoliposomes, and hybrid lipid–polymer nanoparticles:

- **Liposomal Remodeling & Programmable Nanoparticle Sizing:** I routinely formulate phospholipid vesicles (POPC, POPG, DMPC) and co-assemble them with amphiphilic copolymers into monodisperse lipid nanoparticles (SMALPs and VEMALPs) with programmable diameters from **60 nm down to 3 nm** (*BBA-Biomembranes* 2025; *Biophysical Chemistry* 2025) as well as carbon-dot remodeled biomimetic vesicles (*ACS Omega* submitted).
- **Ultra-Mild (<60 s) Self-Assembly for Fragile Payloads:** To prevent denaturation of sensitive biological cargoes caused by organic solvents, high shear, or freeze–thaw cycles, I developed aliphatic **cyclohexyl-VEMA (CH-VEMA)** copolymers that spontaneously

self-assemble with lipid bilayers in **under 60 seconds** in aqueous buffer, maintaining $\sim 10nm$ monodispersity over 72+ hours.

- **Antiviral Biomaterials & Reconstitution of Membrane Targets:** I co-developed cross-linked **network polymers incorporating lipid-bilayer disrupting chains for antiviral applications** (*Polymer Chemistry* 2022) and reconstituted collaborator-purified membrane proteins (**human TRPV1**, **KCNE1**, **KCNE4**, and **gp28**) into native-like lipid bilayers, probing their structural dynamics via **HPLC, SEC/GPC, DLS, TEM, SEM, 1D ^1H/DOSY NMR, and CW-EPR**. These capabilities directly support engineering liposomal nanotraps to capture HIV virions and characterizing biomaterial–cell interactions alongside mammalian cell culture and flow cytometry workflows.

4. Research Leadership, Project Coordination & Laboratory Operations

Throughout my doctoral tenure on multi-PI NIH- and NSF-funded grants, I have served as a lead researcher coordinating experimental workflows between synthetic chemistry and biophysics laboratories. I have **directly mentored 11 undergraduate and 2 graduate students** in experimental design, synthesis, analytical instrumentation, and manuscript writing (yielding **5+ co-authored manuscripts**). Complemented by my tenure as **President of the ACS POLY/PMSE Cincinnati Chapter (2024–2025)** and three years of administrative leadership as an **Assistant Education Officer (2017–2020)**, I bring proven strengths in laboratory organization, procurement, protocol standardization, and guiding junior trainees toward high-impact NIH project milestones.